

Fig. 6: SMD prototyping board/surfboard



Fig. 7: Prototype showcasing

mirror package of the component and/or package conflicts

3. Assembly errors: Wrong component insertion, i.e., a 100-kilo-ohm resistor used in place of a 10-kilo-ohm resistor

Despite the difficulties, electronics prototypes offer many advantages as a distinctive tool in the design process. Working prototypes are the key to understanding a complex interaction between person and object, and advanced techniques make them a more integrated part of the interaction design process.

Prototype showcasing

It is important to understand the difference between a prototype as part of the design process and a prototype as its outcome. In the former case, it serves as a tool for the designer to reflect upon and refine the interface and object in question. In the latter, it becomes a way for the designer to communicate a concept to a broader audience. As exhibition pieces, however, they require the additional time and effort to make them robust and self-reliant.

When it comes to exhibiting (or deploying) electronics designs, the designer needs to move beyond the fragile and delicate breadboard, as it is very difficult to transport breadboard-based prototypes and make them work

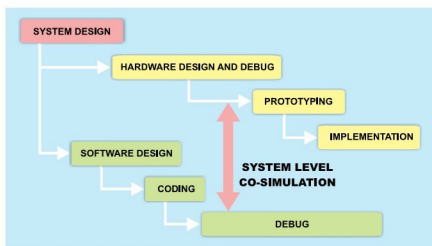


Fig. 8: Block diagram of virtual prototyping environment

for portable use. Another problem with breadboard/generic printed circuit board based prototypes is that they are one-offs: Reproduction is expensive, time-consuming and complicated.

One approach to making prototypes more convenient and flexible is the use of special software such as the Fritzling application for Arduino platform. This allows designers to prepare their electronic inventions for exhibition or production. Simply recreating the tested hardware circuit in software form makes an archivable and shareable file. The shared content includes breadboard, schematics, enclosure and PCB view and may contain additional information such as part ordering numbers. This enables others to interactively scrutinise the electronics design. In effect, the designer creates a robust and long-lasting version of his prototype, close to an end-user product level.

Virtual prototyping?

Designing today's high-end electronic devices is a real challenge, mainly because of the complexity and time-to-market pressure. Most functions of these devices are often grouped into a single integrated circuit, which is then called a system-on-chip (SoC). The design of such systems requires a joint development of custom hardware and software.

The need for faster and better SoC and software development has led to a methodology called 'virtualised systems development.' This approach lets

designers quickly build virtual prototypes comprising software models of system hardware, drawing upon libraries of fast processor and peripheral models. The result is a high-level model of hardware. By running simulations using these models, systems architects and software developers can perform hardware/software partitioning, develop and debug hardware-dependent code, and start applications software development before any system hardware is actually built.

In a conventional product development process, a physical prototype is usually constructed to prove the design concepts, evaluate design alternatives, test product manufacturability or just demonstrate the product. But a virtual prototype (or digital mock-up) is a computer simulation of a physical product and the construction. The testing process of a virtual prototype is called virtual prototyping. In summary, virtual prototyping is nothing but an advanced simulation process.

Virtual prototyping is a relatively new technology which calls for the use of virtual reality and other computer technologies to create digital prototypes. It involves the process of simulating in software the user, the product and their combined (physical) interaction through the different stages of product design, plus the quantitative performance analysis of product. ●

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